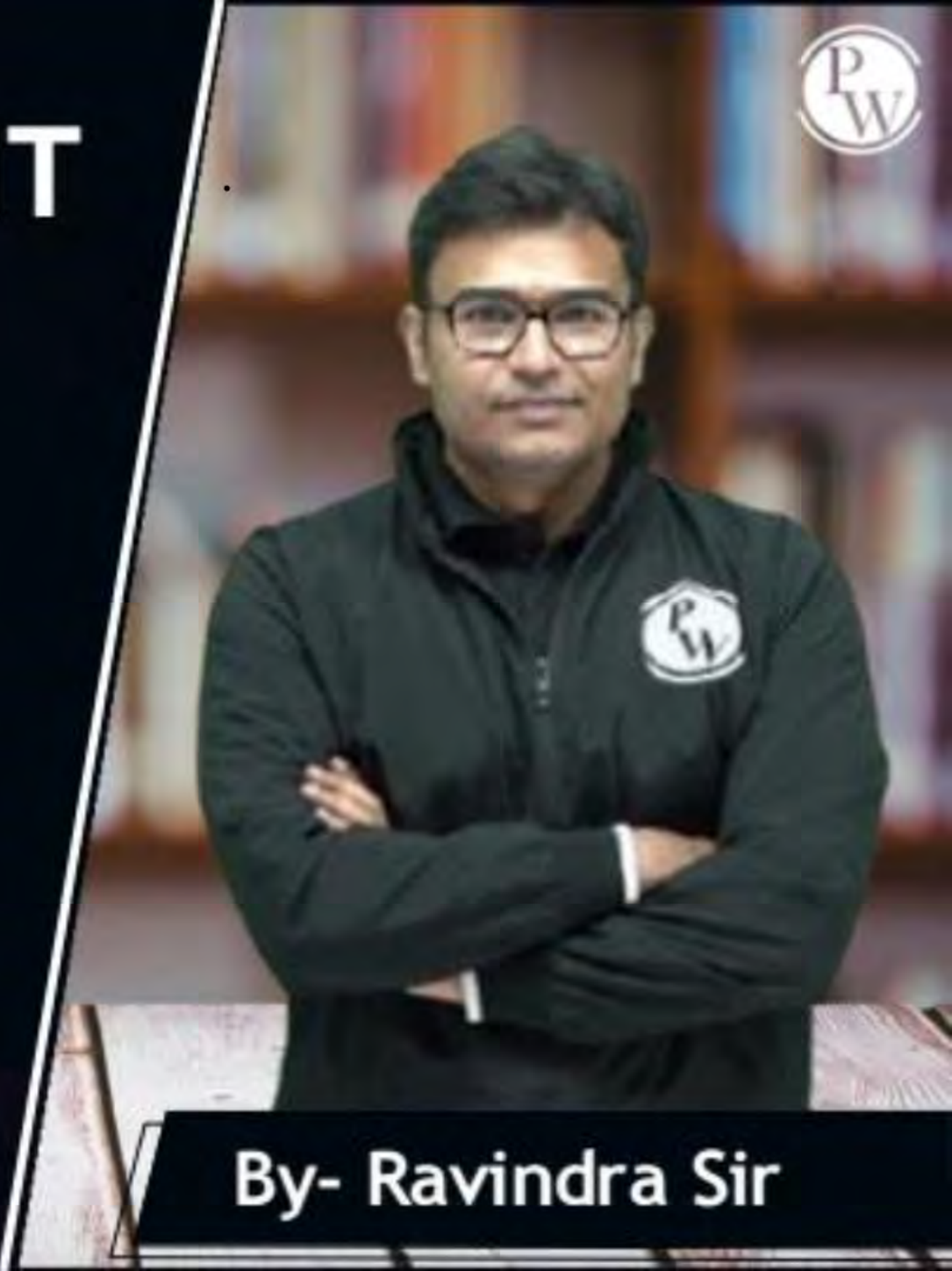
Computer Science & IT

COMPUTER NETWORKS (CN)

Flow Control Methods

Lecture No. 04



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Flow Control

Topics to be Covered



- Topic
- Topic
- Topic

access control



Inspiring Stories : Kunwar Singh

Kunwar Singh – The 80-Year-Old Warrior

Background: A zamindar from Bihar in the 1800s who had never been a soldier.

Struggles: British army at his door, no young age strength, one hand injured badly in battle.

Achievements: Cut off his own wounded hand and kept fighting. Defeated the British in multiple battles during the 1857 revolt.

Impact: If an 80-year-old with one hand could crush a trained army, you can crush a question paper.



Inspiring Stories : Suryasarathi Bose

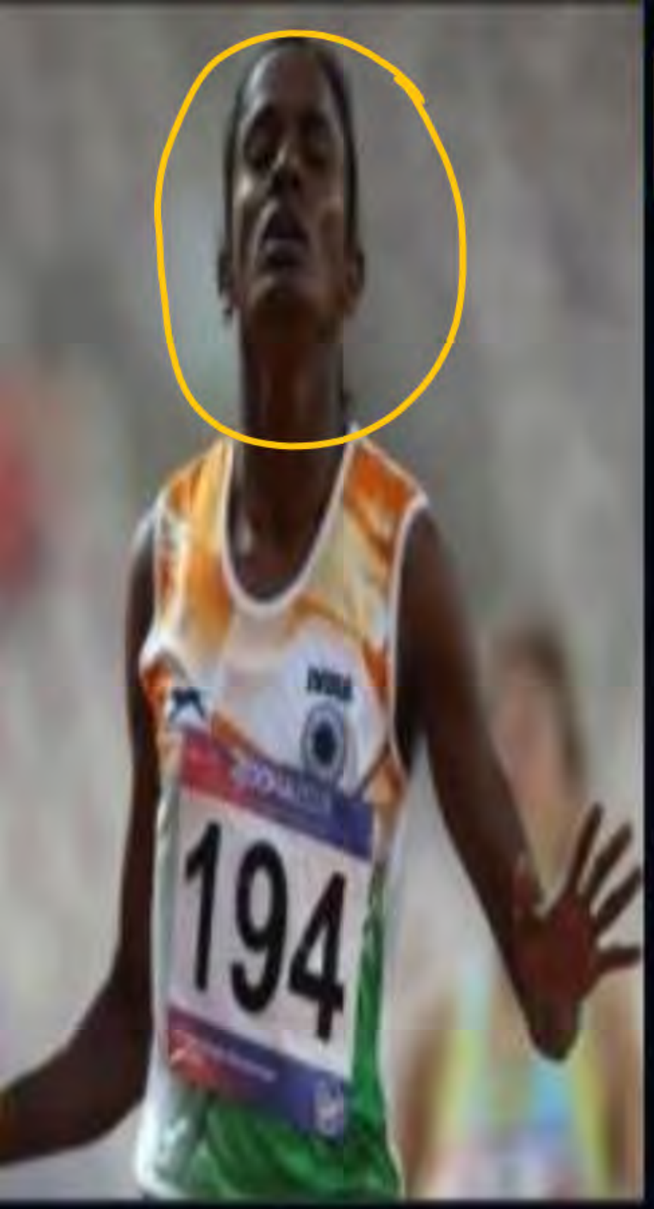
Suryasarathi Bose

Background: Grew up in a simple family with few resources. Not known outside science circles.

Education: PhD in Materials Engineering from IISc Bangalore.

Career Achievements: Made a special water filter using a thin sheet with silver, titanium dioxide, and carbon nanotubes. It kills germs and cleans water fast.

Impact: Gives poor villages and towns a cheap way to get clean drinking water.



Inspiring Stories : Malati Tambe

Malati Tambe – The Unstoppable Runner

Background: Born in rural Maharashtra, worked in fields and as a domestic helper.

Struggles: No shoes, no training, no support. Had to run barefoot in torn clothes.

Achievements: Won multiple state marathons, beating trained athletes. Became a symbol of pure grit.

Impact: If someone can outrun champions without even proper shoes, you can outrun your doubts and win your exam.



Inspiring Stories : G. Mohan Rao

G. Mohan Rao

Background: Came from a modest home. Stayed away from the spotlight.

Education: Advanced degree in Instrumentation Applied Physics from IISc Bangalore.

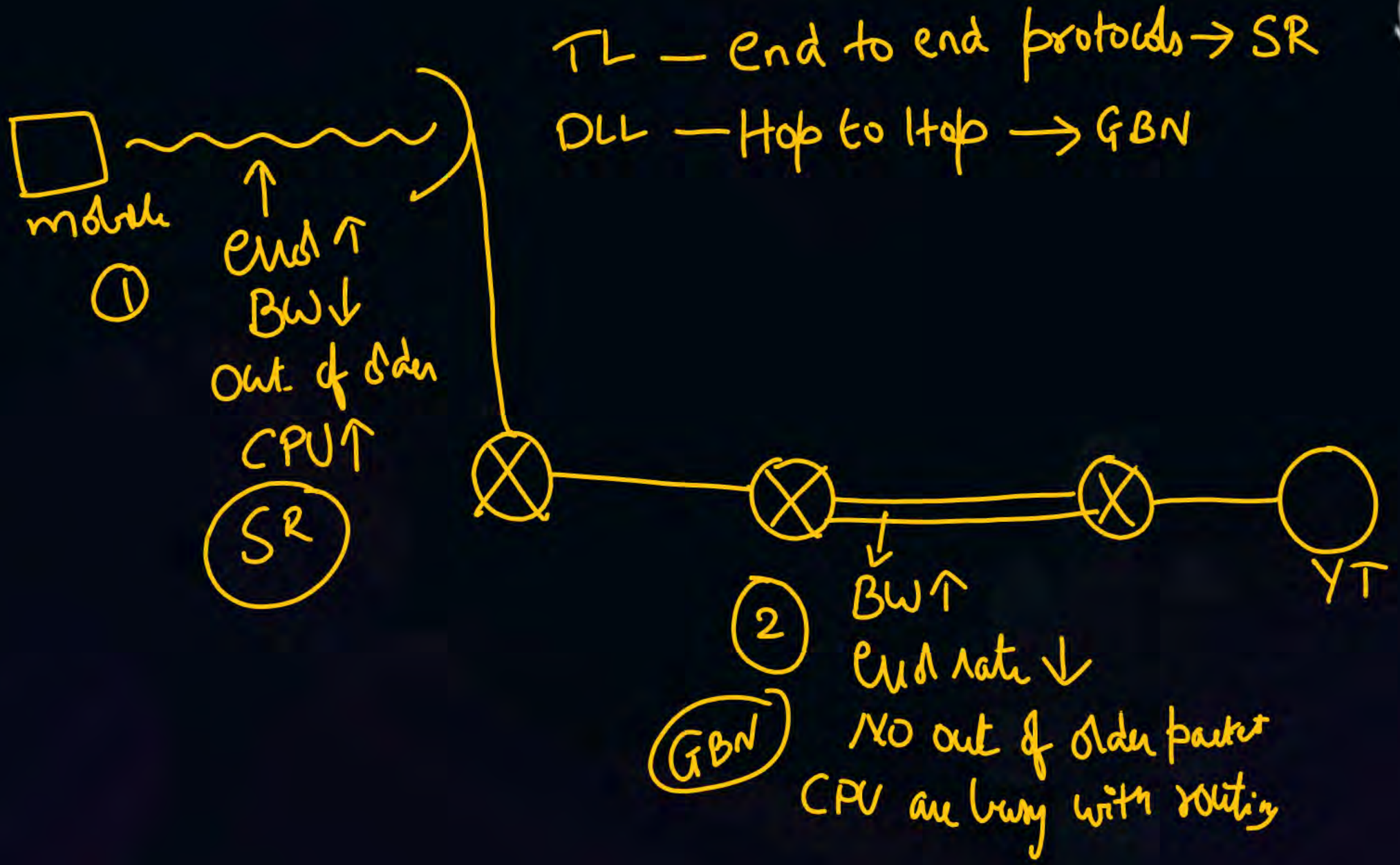
Career Achievements: Created cheap artificial eye lenses (IOLs) with a thin coating. These lenses are now made by AUROLAB.

Impact: Made cataract eye surgery affordable for millions in India and other countries.

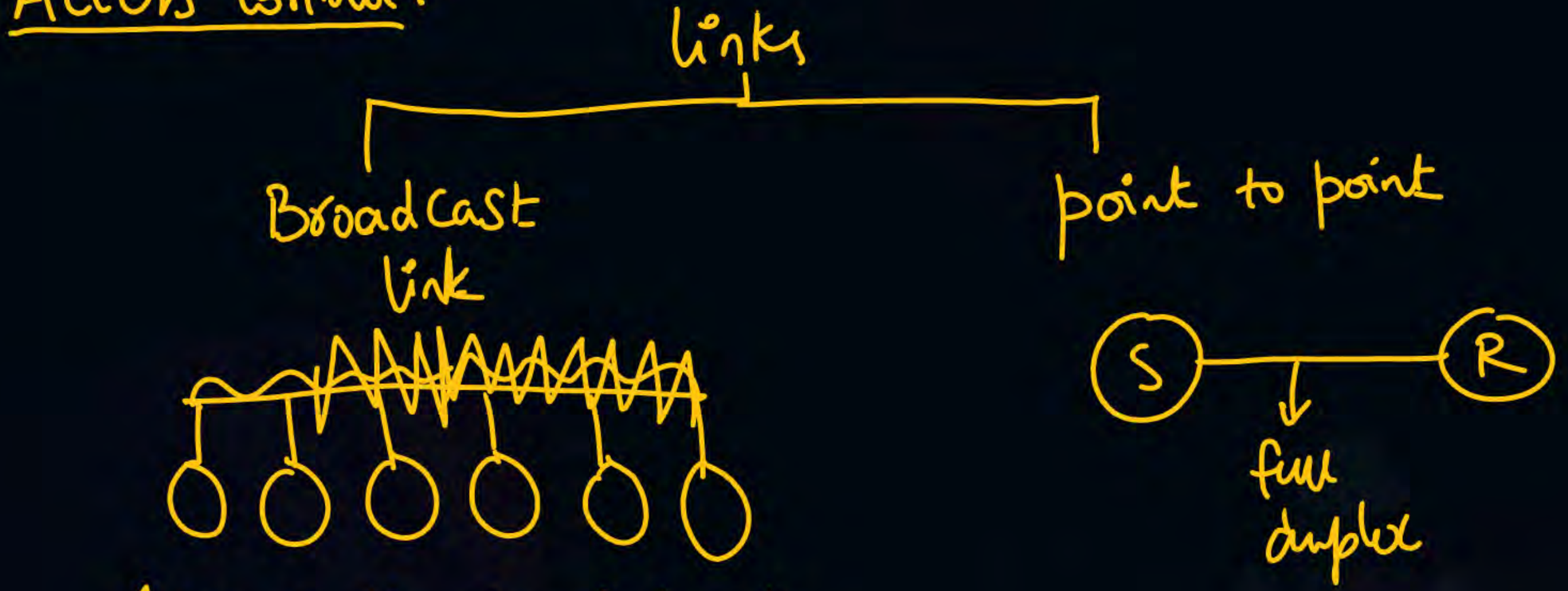


1) BW↑ CPU↓ Buffer↓ (GBN) –

2) BW↓ CPU↑ Buffer↑ (SR) –


TL — end to end protocols → SR
DLL — Hop to Hop → GBN
modem
①
cong ↑
BW ↓
out of order
CPU ↑
SR
BW ↑
cong rate ↓
No out of order packet
CPU are busy with routing
②
GBN
YT

Access Control :-

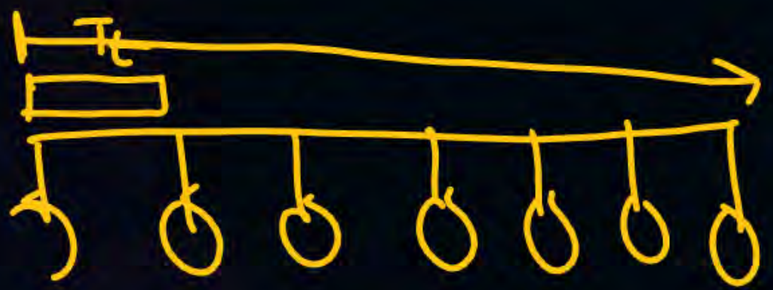


Access to the channel has to be controlled.

Time division multiplexing :-



Cycle time
$= T_t + t_p$



$T_t + T_p$

$$\eta = \frac{\text{useful time}}{\text{cycle time}}$$

$$= \frac{T_t}{T_t + T_p}$$

$$= \frac{1}{1 + \frac{T_p}{T_t}} =$$

$$= \frac{1}{1+a}$$

Time is divided into slots and each machine will get a slot in Round Robin manner.

Time of each slot = $T_t + T_p$ = Cycle time

$$\eta = \frac{\text{useful time}}{\text{Cycle time}} = \frac{T_t}{T_t + T_p} = \frac{1}{1 + T_p/T_t} = \frac{1}{1+a}$$

$T_p = 1ms$, $T_t = 1ms \rightarrow \eta = ?$

$$\eta = \frac{1}{1+a} = \frac{1}{1+1} = 50\%$$

If BW = 4mbps, Throughput = $\eta \times BW = \frac{1}{2} \times 4mbps = 2mbps$

If N stations are connected to the above channel, and each station requires 2 kbps, then how many max no of stations can be connected to the channel?

$$N \times 2\,\text{kbps} = 2\,\text{mbps}$$

$$N = 1000$$

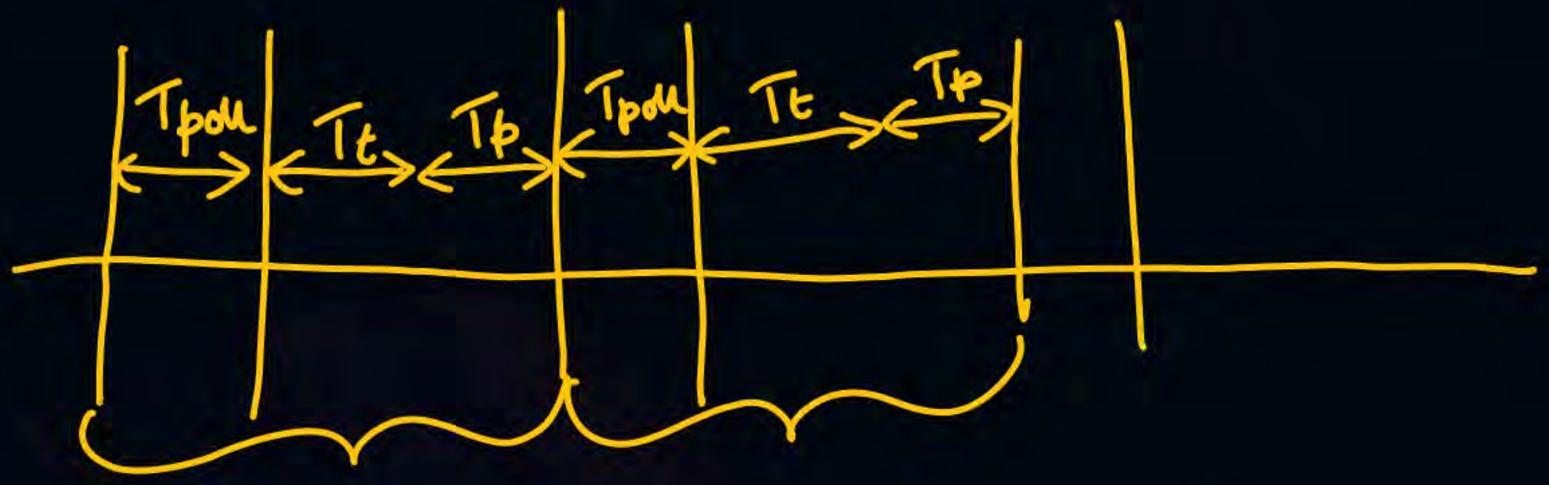
Disadvantage of TDM :-



If a Station is alloted a slot but it doesn't have data to transmit → Then Slot is wasted

Polling :-

Dis :- Polling time



$$\eta = \frac{UT}{CT} = \frac{T_t}{T_{poll} + T_t + T_p}$$

$T_t = 1ms$

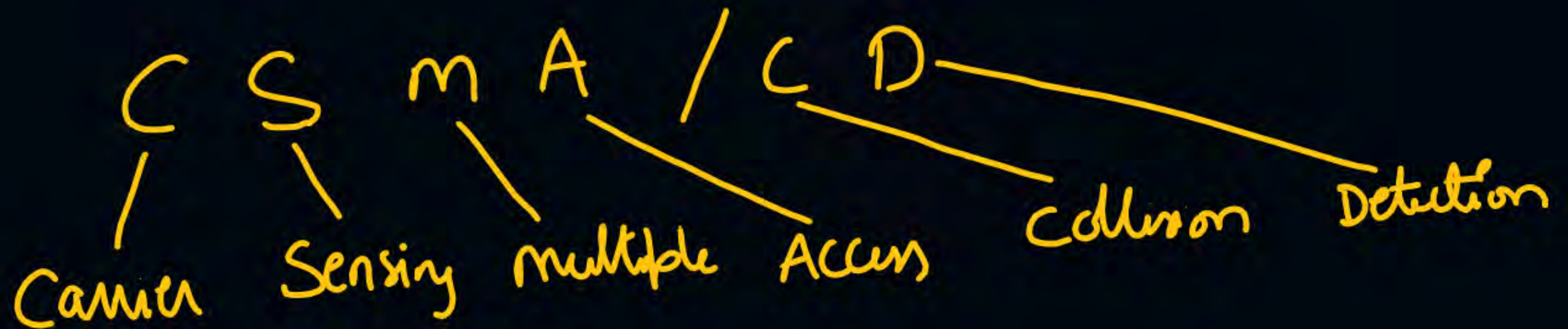
$T_{poll} = 1ms$

$T_p = 1ms$

$$\eta = \frac{1}{1+1+1} = 33.33\% \quad (1/3)$$

$BW = 3mbps$

$Thoughput = \eta \times BW = 1/3 * 3mbps = 1mbps$


C S M A / C D
Carrier
Sensing
multiple
Access
Collision
Detection

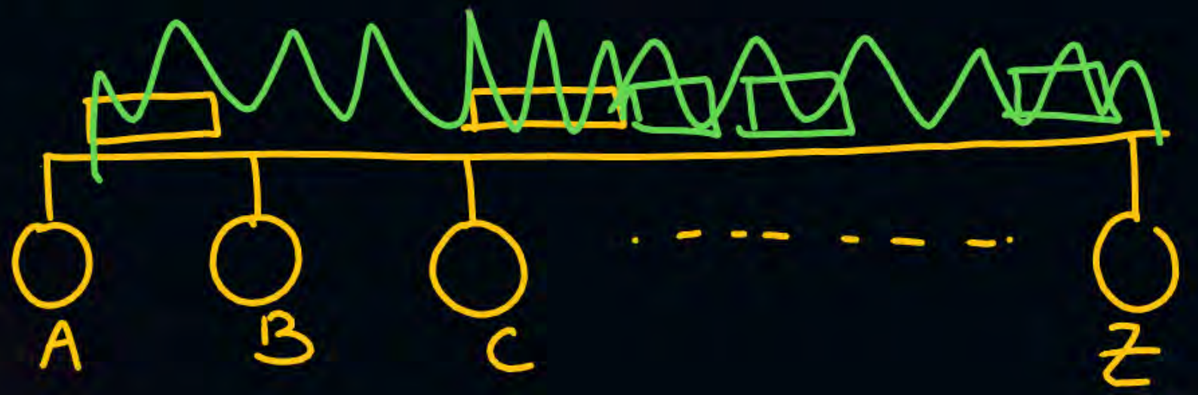
CSMA/CD

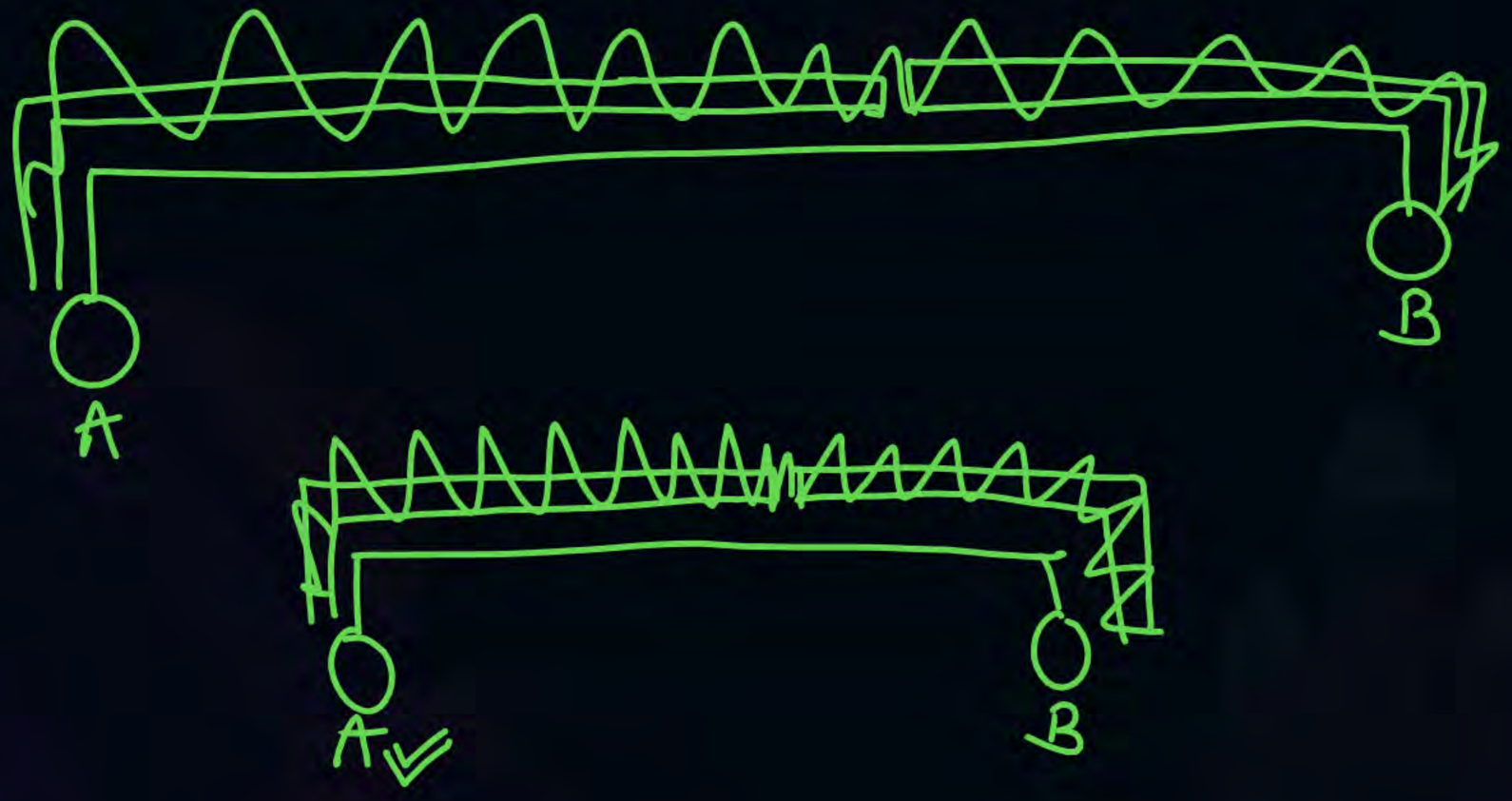


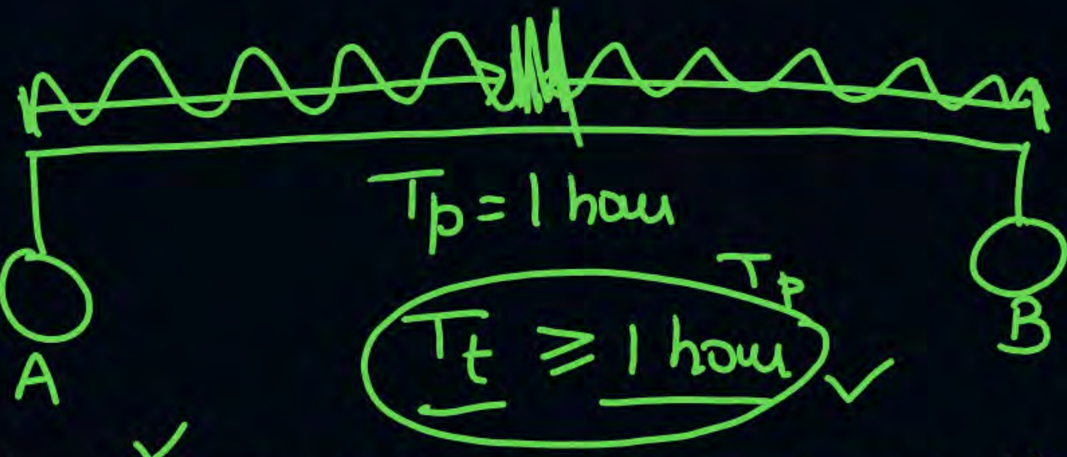
Any station can transmit data anytime. Only condition is Carrier has to be sensed. If carrier is free, we send the data. A Station cannot sense the entire carrier. It can only sense the point of contact.



There are no ack in CSMA/CD



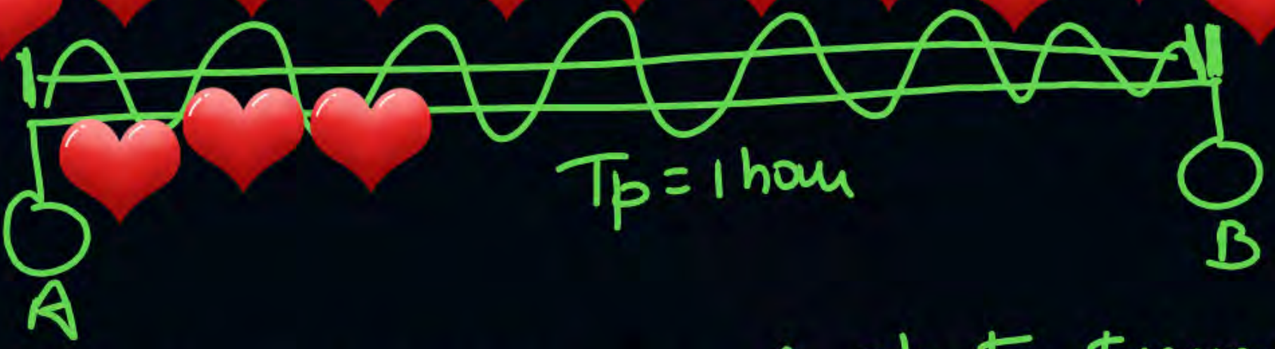

A
B
A
B



Time: 10:00 Am → A, B Senses the Carrier and finds out that Carrier is idle, so Both of them will start transmitting

Time: 10:30 → collision

Time: 11:00 → Both A, B Recieves the collision Signal



At time t = 10:00 AM, A starts transmitting

At time t = 10:59:59

At time 11:00 am ⟶ B starts

At time 12:00 PM → A will receive collision

In worst case, a station will rec collision signal after

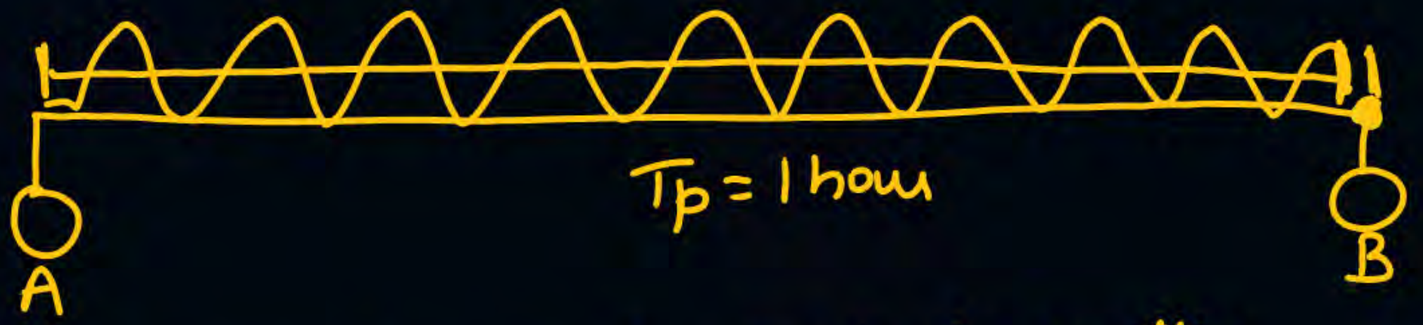
$T_t \geq 2 * T_p$ → collision detection $2 * T_p$

In CSMA/CD, $T_t \geq 2 * T_p$ for collision detection

$$\frac{L}{B} \geq 2 * T_p$$

$$L_{min} \geq 2 * T_p * BW$$

a) Repeat b) No Repeat



Time . 10:00 Am → A starts transmitting

Time 10:59:59:59 → B will sense the carrier

Time 11:00 Am → Collision happens

Time 12:00 Pm → A will get Collision Signal

(Worst Case) $T_t \geq 2 * T_p$ ✓✓ → Collision detection

$\frac{L}{B} \geq 2 * T_p \Rightarrow L \geq 2 * T_p * BW$

In CSMA/CD → For Collision detection

❤ $T_t \geq 2 * T_p$

$\frac{L}{B} \geq 2 * T_p$

$1000 + \frac{1000}{\text{add padding}}$

$L_{min} \geq 2 * T_p * BW$

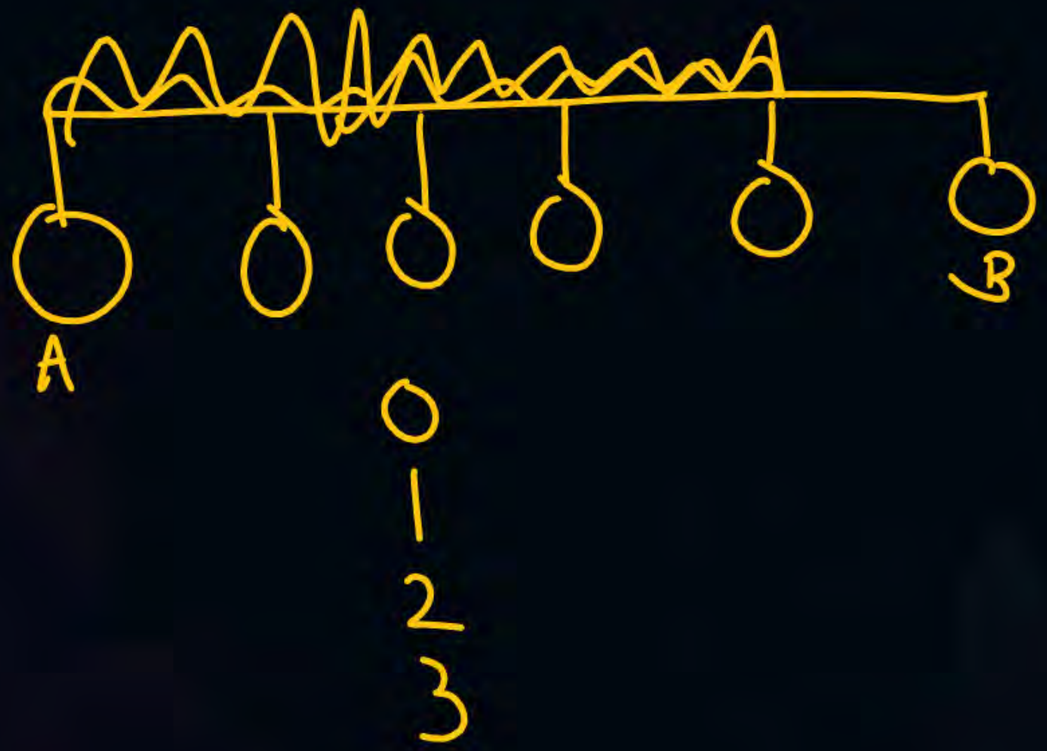
L_{min} → Collision detection

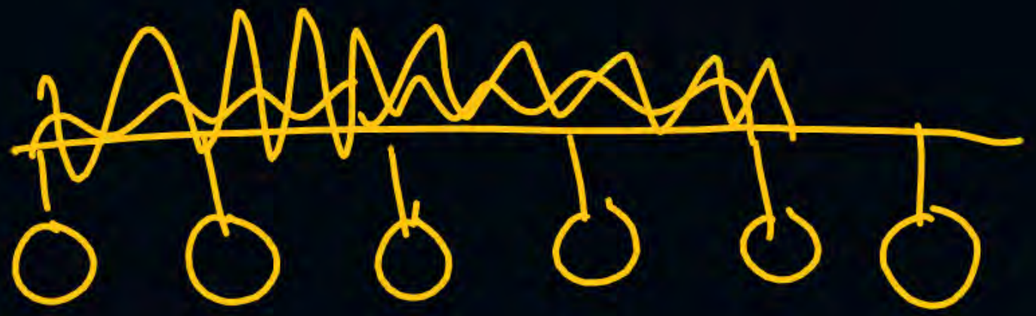
$T_p = 1ms \quad B = 1mbps$

$L_{min} \geq 2 * 10^{-3} \text{ sec} * 10^6 \text{ b/s}$

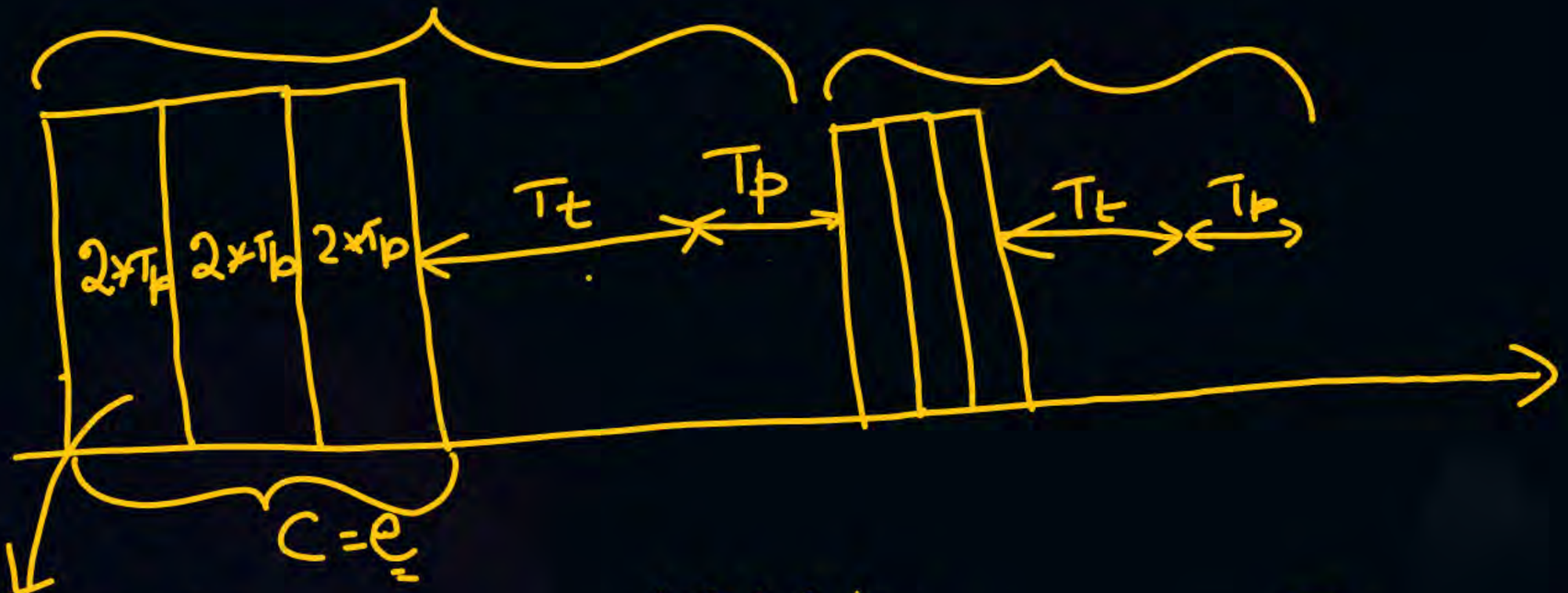
$L_{min} \geq 2000 \text{ bits}$

Efficiency of CSMA/CD :-





0 Collision
1 Coll
2 Coll
3 Coll



Collision slot
Contention slot

$$\eta = \frac{\text{useful time}}{\text{Cycle time}} = \frac{T_t}{(C)^{e} * 2*T_p + T_t + T_p} = \frac{1}{1+6.44a}$$

To derive 'C' we will use probabilistic analysis :-

1) 'N' Stations are Connected to CSMA/CD N/W.
   And every Station wants to send the data with probability 'p'
   There will be a Succesful transmission only if one station transmits the data and (n-1) stations doesnot transmit.

2) Prob for Success is when only one station transmits the data

$$P_{Succ} = {}^{n}C_{1} \; p \; (1-p)^{n-1}$$

$$P_{Succ} = {}^{n}C_1 \times p \times (1-p)^{n-1}$$

what is the max value of P_{Succ}?

$$\frac{d\,P_{Succ}}{dp} = 0 \Rightarrow \boxed{p = 1/n}$$ | P_{Succ} is max

$$P_{max\ Succ} = {}^{n}C_1 \times \frac{1}{n}\left(1-\frac{1}{n}\right)^{n-1} = \left(1-\frac{1}{n}\right)^{n-1}$$

$$\lim_{n\to\infty} P_{max\ Succ} = \lim_{n\to\infty}\left(1-\frac{1}{n}\right)^{n-1} = \frac{1}{e}$$ (L Hospital rule)

→ If $P_{max\ succ} = \frac{1}{e}$, then how many time should we try before getting a success

Poisson distribution :- If a coin is tossed, then if probability of getting head is 'p', then how many time should we toss before getting first head → $\frac{1}{p}$

$\frac{1}{1/e}$ "e"

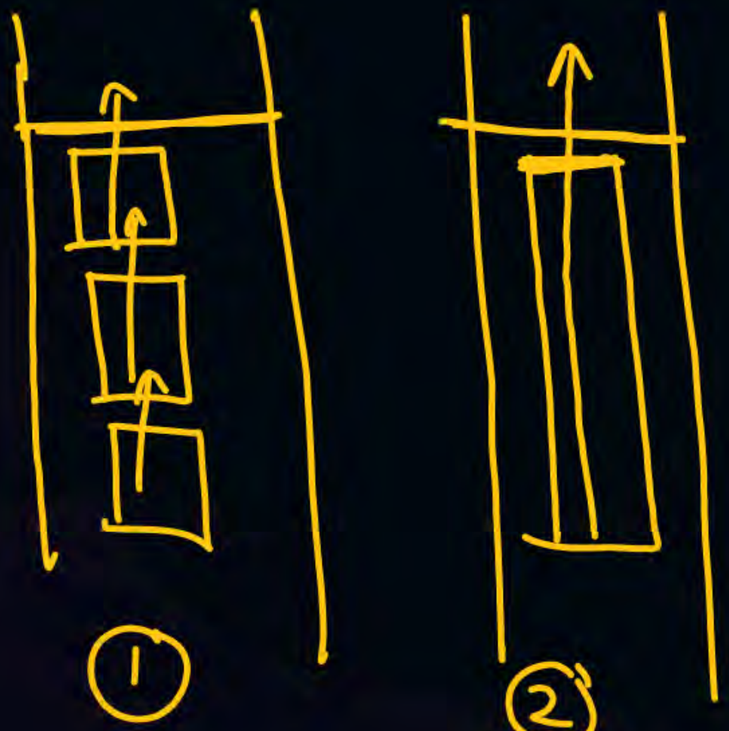


$$\eta = \frac{1}{1+6.44a} = \frac{1}{1+6.44\frac{T_p}{T_t}} = \frac{1}{1+6.44\times\frac{d}{v}*\frac{B}{L}}$$

$L\uparrow \quad \eta\uparrow \rightarrow$ Bursty data

$d\uparrow \quad \eta\downarrow \rightarrow$ LAN





THANK - YOU